ZOO515T SUBJECTIVES

FINAL TERM- CURRENT PAPERS

BIOLOGICAL TECHNIQUES

GUESS PAPER 2024

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Objective:

- Magnification is measured in **scale bar.**
- **Refraction** is the change in direction of a wave passing from one medium to another.
- First analytical ultracentrifuge was developed by Svedberg in **1920.**
- Types of centrifugation **2 .**
- Soxlet extrator was originally designed for the extraction of **lipid ,**
- Chemical kinetics is measured in **uv spectroscopy**
- 1g solute/1000000 , **parts per million**
- Correct sequence of tissue preparation in cryomicrotomy **embedded~sectioning and mounting~freez dryer**
- 1kg is= **1000g.**
- Micropinning is used for **insects.**
- microtome was invented in **1856 Wilhelm His**
- chromatography coined its name in **1943 by Synge and Martin**
- ice point in fahrenheit scale is **32°F**
- the scale accurately used in labs is **analytical scale**
- RT-PCR stands for **reverse transcription polymerase chain reaction.**
- the microscope in which image is seen bright with dark background is called as **dark field microscopy.**
- confocal microscopy was invented in **1957 by Marvin Minsky**
- PCR was invented by **Kary b Mullis.**
- Commonly used used fixative is **formalin**
- Quantitative PCR also called **real time PCR.**
- Most accurate balance is **analytical.**
- **Bio assay** guided fractionation isolate chemical agent from natural origin.
- Lyophilization has **three** steps.
- **Genetic transformation** is the incorporation of **naked DNA** from extracellular environment
- During sectioning with cryostat tissue should be placed for about **30_ 60 min.**
- Steel knives fits the **cryostat.**
- Northern blot used to detect **RNA**
- Radioactive labelled sequence of DNA is called **probe.**

- Magnification can be calculated by **scale bar.**
- **Cryomicrotome** is used for cutting.
- Invention of spectrophotometer in **1940, by Arnold J. Beckman**
- Chromatography is used for **separating mixture.**
- .The length of packed colum gas chromatography **1.5-10m**
  - wave length rays of UV-visible **800-200nm**
- the solution known concentration can be **standard solution.**
- leading strand in DNA is **in continous** manner.
- radio activity known labeled sequence DNA called **probe**
- Oligonucleotide probes are short single strand **DNA fragments**
- result of collection may be effected by
- Most safest and readily killing **ethyl acetate**
- How much agarose is dissolved
- In transformation, DNA is from **extracellular environment.**
- In DNA fingerprinting, **southern blotting** is used
- Ocular micrometer measures the **length, breadth and diameter.**
- **Bioassay-guided fractionation isolate a pure chemical agent from natural origin.**
- **Commonly used fixative is Formalin.**
- **Spectrophotometer is used to measure amount of DNA.**
- **Accurate balances Analytical balances.**
- **Final elongation temperature 70 to 74.**
- **DNA microarray also known as DNA chip or biochip.**
- **There are 2 main types of sampling bouts.**
- Most materials can be dried to **1-5%** residual moisture.
- The science of investigating small objects using a microscope is known as..**microscopy.**
- The word microscope is derived from (**Greek: mikron = small and skopein = to look)**
- The first Microscope is developing in which year…**1590**
- **Giovanni Faber in 1625** coined the name microscope for the compound microscope Galileo submitted to the Accademia dei Lincei.
- Galileo Galilei develops the microscope in **1609**
- How many general parts of a microscope **14 parts**
- **An eyepiece** is that part of an optical system, (in a microscope) which is directed to the viewer.
- An eyepiece converts the real- enlarged-intermediate- image from the objective into **an enlarged- virtual-image**
- The main task of Lens tube in microscope is to **hold eyepiece**
- The diopter adjustment is a **Control knob** on your binocular.
- An ……**Objective lens**……is that part of an optical system, which is directed to the object. It´s task is to collect the light rays,that are reflected from the observed item.
- The…**Clip** ……. serves as a holder for the object plate
- Cross table is a …**Technical term.** .
- There are…**Adjusting screws**……, which can be used to move the table very precisely.
- The screws are provided with a **Measuring scale**
- A …**Condenser ..** is an optical lens which renders a divergent beam from a point source into a parallel or converging beam to illuminate an object
- How many lenses are present in condenser **one or two lense.**
- An ..**Aspheric lens**…lens avoids chromatic aberration
- With the …**fine focus..**one can regulate the distance between object and objective
- The fine focus moves the stage only…**minimally**
- Like the fine focus, the …**course focus..**also moves the stage to regulate the difference between object and objective
- The early microscopes used …**concave**…mirrors to refrect light on the objects.
- Most microscopes operate with…**LED**…light.
- The base of microscope guarantees the necessary…**stability**……to the microscope.

- Luminous-field diaphragm can be used to **Reduce the illumination and adapt it to the size of the object**
- How many types of microscope …**two**
- **Simple light microscopes** use a single lens to magnify an object and cannot reach high magnification
- Compound light microscope has a higher…**magnification**…power
- The simplest microscope of all is a **Magnifying glass**
- Microscopes used in homes, schools, and professional laboratories are actually compound microscopes and use at least two lenses to produce a magnified image **true**
- Most compound microscopes can magnify by…**10,20,40……, or …100**…… times.
- MAGNIFICATION is defined as…**Degree of enlargement**
- Magnification is depends upon how many factors **3 (Optical tube length 2. Focal length of objective 3. Magnifying power of eye piece**
- the ……**Refractive index.** …..which is a measure of how great a substance slows the speed of light
- **Refraction**…is the change in direction of a wave passing from one medium to another
- …**Refractive index..**also called index of refraction, measure of the bending of a ray of light when passing from one medium into another.
- In microscopy, the term **…Resolution**….is used to describe the ability of a microscope to distinguish details of given specimen. This is the resolving power of microscope.
- …**Diffraction**..refers to various phenomena that occur when a wave encounters an obstacle or opening
- …**Wavelength of light (λ)** …is used to examine a specimen
- The smaller is the wavelength of light (λ), the greater is its ability to **resolve the points on the object**
- The …**smaller**….is the wavelength of light, the greater is its resolving power
- …**Numerical aperture….**is a number that expresses the ability of a lens to resolve fine details in an object being observed.
- Light diffracted by the specimen is presented as **an inverted cone of half-angle (α)**
- Higher values of numerical aperture permit increasingly oblique rays to **enter the objective front lens**
- Most objectives in the magnification range between ……**60X…..**and ……**100X**….. (And higher) are designed for use with immersion oil.
- …**aperture** ..can be defined as the opening in a lens through which light passes to enter the camera
- Aperture It is expressed in **F numbers.**
- **Focal length** ….. is the distance at which a beam of light will be focused to a single spot
- **Bright field illumination**…is useful for samples which have an intrinsic color.
- …**Staining** ..is often required to increase contrast, which prevents use of live cells in many situations.
- Bright-field illumination is useful for samples that have an intrinsic color, for example …**RBCs in animal cells**
- The practical limit to magnification with a light microscope is around **1300X**
- The microscopy, where only the specimen is lit by a light or electron beam, and the rest of the specimen field is dark is known as **Dark field microscopy**
- Phase contrast microscopy first described in… **1934**
- Who was awarded Nobel Prize in physics in **1953 Fritz Zernike**
- Phase-contrast microscopy is particularly important in… **Biology.**
- Phase-contrast microscopy is particularly important in biology and It reveals many **Cellular structures**
- Phase Contrast Microscope based on the……**Wavelength (nature) of light rays**
- Different shade of grey are distinguished to our eyes Principle Phase Contrast Microscope due t o …**Differences in amplitude of light rays**

- … **PCM** … converts invisible small phase changes caused by the cell component in to visible intensity changes.
- When diffracted and surrounding waves do not over lap, this condition is known as **Out of phase**
- In a Phase contrast microscope, one set of light rays comes directly from the **light sources s. waves**
- Light passing through one material & into another material of slightly different refractive index or thickness will undergo a change in **phase**
- Phase contrast is obtained with the help of the **phase ring**
- … **Refractive index (μ)..**is the ratio of Velocity of light in air to that of in given medium.
- How many types of Phase contrast microscopy **Two main (positive and negative)**
- Light can be considered a form of wave motion consisting of **sinusoidal waves**
- …… **Electron microscopy (EM)** ….is a technique for obtaining high resolution images of biological and non-biological specimens.
- The high resolution of EM images results from the use of **electrons**
- … **EM images**..provide key information on the structural basis of cell function and of cell disease.
- In … **1926 HANS BUSCH** ….developed the electromagnetic lens.
- The first prototype electron microscope, capable of four-hundredpower magnification, was developed in 1931 by the physicist …… **ERNST RUSKA & MAX KNOLL**…..and the electrical engineer
- … **Siemens** … produced the first commercial electron microscope in 1938
-)How many types of electron microscope **2 Transmission EM (TEM), Scanning EM (SEM).**
- The very best light microscopes have a resolution limit of about **0.2 μm,**
- Viruses are too small to be seen with **light microscopes**
- The probability of electron scattering is determined by the… **density (atomic number)** … of the specimen atoms
- Two other important techniques for preparing specimens are **negative staining and shadowing**
- In shadowing, a specimen is coated with a thin film of platinum or other heavy metal by evaporation at an angle of about … 45° ……. from horizontal
- The key feature of fluorescent microscopy is that it employs reflected rather than … **transmitted light**
- … **Optical filters..**include a set of a compatible excitation filter, emission filter, and dichroic beam splitter; – An excitation filter selects the wavelengths to excite a particular dye within the specimen.
- (TIRFM) Stands for **Total internal reflection fluorescence microscopy**
- … **Beam splitter**...is an optical device that splits a beam of light in two.
- A xenon arc lamp is a highly specialized type of **GAS discharge lamp**
- Sigma and pi bonds are chemical… **covalent bonds**
- Sigma bonds are formed by**…end-to-end overlapping**
- sigma bonds are…… **stronger** ….. than pi bonds.
- To observe the sample through a fluorescence microscope, it should be first labeled with a fluorescent dyes/substance known as a **fluorophore.**
- The emission spectrum is usually … **sharper** …….than the excitation spectrum.
- **Fluorescence microscopy**….is a critical tool for academic and pharmaceutical research, pathology, and clinical medicine
- Fluorescence microscope is used to tracing the location of a specific … **protein** ……in the specimen
- Confocal microscopy was pioneered by … **Marvin Minsky..**in 1955.
- … **Micrometry** ….is the science in which we have some measurement of the dimensions of an object being observed under the microscope.

- How many Types of Micrometers **Two (stage micrometer and 2. ocular meter or ocular micrometer**
- … **calibration** …the comparison of measurement values delivered by a device under test with those of a calibration standard of known accuracy
- It is the process of preserving the tissues in the natural condition **Fixation**
- In fixation, poorly fixed specimens are more**…difficult**….. to section than the well fixed ones
- commonly used fixatives are **alcohol, formalin, glutaraldehyde**
- the commonly used fixatives **10**% formalin or **2.5% glutaraldehyde** is used
- fixatives should be ……**Non-toxic**…and non-allergenic
- commonly used Coagulant fixative are ……**50-60%….**ethanol, 80% methanol.
- picric acid and trichloroacetic acid are **Acidic coagulants**
- …**Trichloroacetic acid (Cl3CCOOH)..**can penetrate hydrophobic domains of proteins
- Bouin solution is a preparation of 5%.....**acetic acid,.....,** 9% formaldehyde, and 1.5% picric acid in aqueous solution
- a **…mutagen**…is a physical or chemical agent that changes the genetic material, usually DNA, of an organism and thus increases the frequency of mutations above the natural background level.
- …**explosive charge**..is a measured quantity of explosive material, which may either be composed solely of one ingredient or be a mixture containing at least two substances.
- An …**explosive (or explosive material)…**is a reactive substance that contains a great amount of potential energy that can produce an explosion if released suddenly,
- **Formaldehyde**…is a Powerful reducing agent
- **Para formaldehyde**…is also commonly used and will depolymerise back to formalin when heated,
- The formaldehyde…**vapor**….. can also be used as a fixative for cell smears
- **mercurials** …penetrate poorly and produce tissue shrinkage.
- Zenker's fixative is a **Mercurials such as**
- Zenker's fixative is named after…**Konrad Zenker** …, a German histologist, who died in 1894
-) Fixation methods can be grouped in …**Two. Physical methods & chemical methods.** …numbered.
- …**Quick freezing**..is convenient to process sample sizes not larger than 2 mm so that freezing is homogeneous through all the tissue thickness, including the inner parts.
- …**Cryoprotection**..is always a good practice in quick freezing
- **Microwave**…is a good heater because temperature rises quite homogeneously and immediately through the sample
- in immersion, volume of the fixative needs to be …**10 to 20 times larger**…..than the volume of the sample.
- common duration for most of the fixatives is …**24 hours**
- in Perfusion, The fixative solution is introduced through**…the vascular system.**
- How many Types of Knives? **DIAMOND♣ GLASS KNIVES ♣Four: (STEEL KNIVES SAPPHIRE KNIVES)♣KNIVES**
- **Steel** blades are used to prepare sections of animal or plant tissues for light microscopy histology.
- **Glass knives** are used in an ultramicrotome to cut ultrathin slices of samples for electron and light microscope applications.
- A diamond knife is a very sharp knife in which the edge is made from diamond, invented by Humberto FernándezMorán in …**1955**
- Micron settings for section thickness can range from …**1. to 60….**microns on most microtomes.
- …**sectioning** ….requires great care as tissues of the diagnostic importance can easily be lost or the block surface may be damaged
- the term “microtome” was coined in 1839 by…….**.Chevalier**

- Microtomy and …**Paraffin Section Preparation…** is a great training aid for new microtomists and is an excellent refresher for experienced operators
- Blade is fixed at …**horizontal……**position in rotary microtone
- cryomicrotome is used for cutting…**frozen tissues.**
- an …**microtome**…..arranged in a cryostat and used to prepare thin sections of frozen tissue for microscopic examination.
- in cryomicrotome, the temperature inside the cryostat is about **…-20° to -30°…….**Celsius
- the…**cryostat..** is the instrument that has the arrangement to freeze the tissue and also to cut the frozen tissue for microscopic section
- The firm ice within the tissue acts as …**embedding media** ….to Tissue Preparation for cryomicrotome
- **Embedding medium** …medium is used to, hold the tissue over the chuck in cryomicrotomy
- Presently …**optimum cutting temperature (OCT) compound** …..(Tissue-Tek) is used as embedding medium in cryomicrotomy
- (OCT) stands for **optimum cutting temperature (OCT) compound**
- A …**vibratome**…is an instrument used to cut thin slices of material (although, usually thicker slices than those cut in paraffin embedded samples using a microtome)
- the morphology of neuron dendritic trees is difficult to image in thin sections (4 to 10 μm), and we may also need to visualize dendrites after …**Immunohistochemistry.**
- another feature of the vibratome is that all the cutting is done in an aqueous solution, usually a …**buffer or saline**…..solution
- The fluorescent intercalating dye used in Electrophoresis is **ethidium bromide** :
- **Polyacrylamide** …. gels are most effective for separating small fragments of DNA (5-500 bp)
- age of Polyacrylamide run in a vertical configuration in a constant **electric field**
-) DNAs from …**50**…..bp to several mega bases in length can be separated on agarose gels of various concentrations and configurations
- As size of DNA fragment increases the percentage of gel composition have to be **decrease**
- agarose gels cast with **low concentrations** ……of agarose (0.1-0.2% w/v) are capable of resolving extremely large DNA molecules.
- In 1984, when Schwartz and Cantor developed **pulsed-field gel electrophoresis (PFGE)**
- commercially prepared agarose polymers are believed to contain galactose residues per chain. **800**
- Agarose is not : : **homogeneous:**
- Lower grades of agarose may be contaminated with other polysaccharides, **salts and proteins**
- Molecules of double stranded DNA migrate through gel matrices at rates that are inversely Rate of migration of DNA proportional to the …**log10**…..of the number of base pairs
- Intercalation of …**Ethidium Bromide**….causes a decrease in the negative charges of the double standard DNA and an increase in both its stiffness and Rate of migration of DNA length
    - obtain maximum Rate of migration of DNA resolution of DNA fragments **…>2 kb** … kb in size, agarose gels should be run at no more than 5- 8 V/cm.
- Name the buffers used in electrophoresis of native, doublestranded DNA **Tris-acetate and EDTA (pH 8.0; TAE) (also called E buffer),**
- double standard linear DNA fragments migrate 10% faster through TAE than through …**TBE or TPE**
- Ethidium bromide is also called as…**6x Gel-loading buffer**
- Power supply device capable of up to……**500 V and 200 mA.**

- in gel electrophoresis voltage is **Apply a voltage of 1-5 V/cm (measured as the distance between the positive and negative electrodes).**
- If ethidium bromide is present in the gel and electrophoresis buffer, examine the gel **byUV light** ….light
- …: **Ethidium Bromide**…is a fluorescent dye ,most convenient and commonly used ,contains a tricyclic Detection of DNA in Agarose gel planar group that intercalates between the stacked bases of DNA.
- UV radiation at **…254 nm**… is absorbed by the DNA and transmitted to the dye and at 302 nm and 366 nm is absorbed by Detection of DNA in Agarose gel the bound dye itself.
- …**Ethidium Bromide**…can be used to detect both single- and double-stranded nucleic acids (both DNA and RNA
- Ethidium Bromide inhibits Detection of polymerization of the **acrylamide**
- electrophoretic mobility of linear dsDNA reduced by…**15… %** in the presence of the dye
- the fluorescence enhancement is **…> 1000**.. times greater than equivalent Ethidium Bromide DNA complex.
-) Photography is carried out with …**green or yellow** ….filters
- Most commercially available devices (transilluminators) emit UV light at **302 nm**
- The instrument that used to cut the exteremly thin sections 40nm 500nm is called **ultra microtome.**
- Which knife is invented by Humberto **diamond knife**.
- An aspheric lense avoids **chromatic aberration**
- Fluorescence microscope additionally require **fluorescence stain**
- Cryostate can be cleaned through **alcohol**.
- Which of the following is the type of embedded medium **celloidin, resin, paraffin wax**
- Wheel action is translated into back and forth movement in which one.. **rotary microtome**
- Microtomes are used in **microscope** allowing for the preparation of sample for observation.
- Phase contrast microscopy reveals both **tissue and cellular** structure.
- Zenkers fixatives employed to prepare of specimen **animal tissue and microbial cell** for microscopic study.
- In microtomy method , during sample collection, the tissue pieces from hollow organs like intestine, oviduct ect should be cut in **transversly**.
- In vibrating microtome, cut thickness is usually around **30-500µm** for live tissue.
- The picric acid in Bouin solution present challenges for safe handling and disposal due to its **mutagenic and explosive** potentials.
- In tissue processing if transfer of specimens or changing reagents is done by hand, this method is called **manual processing**.
- Fixation is the preservation of the biological **tissues** from decay due to autolysis and putrefaction.
- Zenker fixation contains potassium **dichromate, acetic acid, water.**
- the fast freezing chemical **Liquid nitrogen**.
- Taq polymerase is **heat tolerant** enzyme .
- The processing of two complementary single stranded DNA or RNA molecules and allowing them to form a double stranded molecule through base pairing is called **hybridization**.
- Sanger sequencing method is used for **DNA**.
- The primer which is the mixture of random hexamer or olido_dt primer is called **non sequencing specific primer**.
- The enzyme use in PCR **DNA polymerase**.
- Sanger sequencing involve making many copies of a target DNA region of **polymerase**.
- The basic steps of PCR is **3.**

- Scientists used DNA **microarrays** to measure the expression level of large numbers of genes simultaneously or to genotype multiple regions of genome.
- PCR method emplify DNA fragments of **0.1 to 10kb**
- Which of the following type of PCR was firstly used for diagnosis of Duchenne mascular dystrophy. **Multiplex PCR**
- A common PCR used in laboratories is an important tool that a can be used to **amplify a sample of DNA.**
- Sanger sequencing method was developed in **1977.**
- Which one is not the step of PCR **…sterilization**
- Taq polymerase an enzyme originally isolated from ***Thermus aquaticus***
- PCR invented in **1983**

Subjective:

1= Difference between molarity, molality, normality.

Molarity; the number of moles of solute (the material dissolved) per liter of solution.
Normality : The strength of solution measured in terms of gram equivalent per litre is called normality. It is denoted by N.
N1V1 = N2V2
Molality : The total moles of a solute contained in a kilogram of a solvent.

2= MaxamGilbert sequencing.

Method:
• Maxam-Gilbert method of DNA Sequencing • It is a method by which the sequence of a DNA fragment is identified by using chemicals, thats cut DNA at specific points. • Also called Chemical degradation method of DNA sequencing.
Procedure
it is a fragment of double stranded DNA, and you do not its sequence:
Step :1 • As the sequence of both strands are unknown, but if we find out the sequence of one strand, we would get to know sequence of other one also. At first, the double stranded fragment is separeted into two single strands by applying high Temperature or high PH.
Step:2 • Run the single stranded fragments on gel. As lighter fragment band will move further than the heavy fragment band. • How will we know which one is the lighter band? • The band having larger number of purines(A,G) would be heavier. • Single stranded DNA fragments in Gel Electrophoresis gel.
Step: 3 • Take one of fragment band from the gel. • Remove the Phosphate at 5′ end and incorporate Radioactive Phosphate 32- PO4 enztmatically.
Step: 4 • Radioactive Labelling • Now put all the radioactively labelled fragments in four tubes. 1 2 3 4
Step: 5 • Chemical Degradation Tube • Tube:1 : Increase Temperature and PH(by adding NAOH), that would cause fragments to break down. Dimethyl sulfate will be added that would make cuts at Adenine and Guanine positions. • Tube 2: Dimethyl sulfate and dilute HCL will be added that would cuts the fragment at Adenine position • Tube 3: Reagents Hydrazine and Piperidine are added that would cuts the fragment at position Cytocine and Thyamine. • Tube 4: In the last tube, Hydrazine, Piperdine and NACL is added that would cuts the fragment at Cytocine position. Dimethylsulphate + High Temp + NAOH Dimethylsulphate + dil HCL Hydrazine + Piperidine Hydrazine + NACL + Piperidine
Step:6 • Take one of fragment band from the gel. • Remove the Phosphate at 5′ end and incorporate Radioactive Phosphate 32- PO4 enztmatically.

1. Traditional cytometry and acoustic cytometry difference

Traditional flow cytometers

The traditional cytometers are the common cytometer using sheath fluid for focusing the sample stream. The most common lasers used in traditional flow cytometers are 488 nm (blue), 405nm (violet), 532nm (green), 552nm (green), 561 nm (green-yellow), 640 nm (red) and 355 nm (ultraviolet).

. . **Acoustic Focusing Cytometers**

In these cytometers, ultrasonic waves are used to focus the cells for analysis. This prevents sample clogging and also allows higher sample inputs.

2. Blood cell types and their functions.

Mononuclear leukocytes

• Mononuclear leukocytes are the only cell type in blood capable of growth • They can be cryopreserved for the establishment of cell lines. • Cryopreservation permits cell viability and can be the only source to measure RNA • single macrophage (monocyte) surrounded by several lymphocytes

Granulocytes

• Granulocytes can serve as a source of DNA without sacrificing the lymphocytes

Erythrocytes (RBC)

• Stored after washing with physical saline • Can be useful to study adducts of haemoglobin **Plasma and Serum**

• Can be used to measure microanalytes, diet components, vitamins, xenobiotic exposures and so on. • Plasma can be obtained from an anticoagulated blood sample through separation from cell components. • Serum is better for antibody measurements, nutrients, etc.

5: Sedimentation coefficient and its formula.

Centrifugation separates particles in a suspension based on differences in size, shape and density that together define their sedimentation coefficient.

S=dr/dt (1/ω2 r)

This is known as the Svedberg equation and is usually expressed in Svedberg units,

S (= 10-13 second).

This equation indicates that ‘S’ is dependent upon the molecular weight, the density and the frictional coefficient.

6: Use of DNA arrays and its types.

• Many types of arrays exist and the broadest distinction is whether they are spatially arranged on a surface or on coded beads: • The traditional solid-phase array is a collection of orderly microscopic "spots", called features, each with thousands of identical and specific probes attached to a solid surface, such as glass, plastic or silicon biochip (commonly known as a genome chip, DNA chip or gene array). Thousands of these features can be placed in known locations on a single DNA microarray. • The alternative bead array is a collection of microscopic polystyrene beads, each with a specific probe and a ratio of two or more dyes, which do not interfere with the fluorescent dyes used on the target sequence. • Many types of arrays exist and the broadest distinction is whether they are spatially arranged on a surface or on coded beads: • The traditional solid-phase array is a collection of orderly microscopic "spots", called features, each with thousands of identical and specific probes attached to a solid surface, such as glass, plastic or silicon biochip (commonly known as a genome chip, DNA chip or gene array). Thousands of these features can be placed in known locations on a single DNA microarray. • The alternative bead array is a collection of microscopic polystyrene beads, each with a specific

probe and a ratio of two or more dyes, which do not interfere with the fluorescent dyes used on the target sequence.

7: Use of spectrophotometer.

- **A spectrophotometer is used for the quantitation of nucleic acid (DNA or RNA).**
- **It determines the concentration of color and colorless compounds by measuring the absorbance of the solution. It can also determine the concentration of biological materials like nucleic acid and proteins.**
- **Similarly, it determines the phase of reaction by measuring the formation and disappearance rate of the light-absorbing compounds in the range of the visible and UV region of the electromagnetic spectrum.**
- **It also identifies compounds by determining the absorption spectrum in the visible region of the light and the UV region of the electromagnetic spectrum.**

8: 5 steps of preservation.

• **Euthanizing.** Specimens should be euthanized in a way that will leave them undamaged and relaxed.

• **Injection and slitting.** Liquid preservatives must be introduced into the body cavity, limbs and tail, either by hypodermic injection or through slits.

• **Fixing.** While the specimens are still relaxed, they should be arranged in trays so that they will harden in the proper position.

• **Labeling**. Each specimen should be accompanied by certain data, either attached directly or entered in a notebook with a number corresponding to a numbered tag tied to the specimen.

• **Storage.** After specimens have been fixed in the proper position, they should be stored in liquid preservative for at least several days, after which they may be allowed to remain in the liquid, or transferred to plastic bags for temporary storage.

9: Cryomicrotome and its significance.

• For rapid medical diagnosis when it is necessary to perform a rapid analysis of a sample, a frozen section is required • The piece(s) of tissue to be studied are snap frozen in a cold liquid or a cold environment (-20° to -70° Celsius).• Freezing makes the tissue solid enough to section with a microtome.• Frozen sections are performed with an instrument called a cryostat, The temperature inside the cryostat is about -20° to -30° Celsius. • The tissue sections are cut and picked up on a glass slide. • The sections are then ready for staining. • If not immediately required.. They should be kept in freezer at -20 C.

10: Note on spectrophotometer

A spectrophotometer is employed to measure the amount of light that a sample absorbs. The instrument operates by passing a beam of light through a sample and measuring the intensity of light reaching a detector.

History of Spectrophotometer

In the 1930s, vitamin research indicated that several vitamins, particularly vitamin A, absorb ultraviolet (UV) light. Spurred by the American government's interest in measuring vitamin content in soldiers' rations using ultraviolet and visible (UV-Vis) light, this research culminated in the commercial launch of UV-Vis spectrophotometers in the early 1940s. The spectrophotometer was invented in 1940, by Arnold J. Beckman and his colleagues at National

Technologies Laboratories, the company Beckman had started in 1935. They were led by project leader Howard H. Cary. The spectrophotometer was the company's greatest discovery.

Types of Spectrophotometery

1 **Single Beam Spectrophotometer**

This is an instrument composed of a Light source, a Wavelength selector (monochromator), a Sample compartment, and a Detector. The type of light source will depend on the desired wavelength. The wavelength selector splits the light spectrum into its component colors and selects a narrow band of this spectrum. Either a prism or a grating may be used as a dispersing element to split the light. The wavelength of the selected light may be adjusted by changing the angle of the dispersing element, so that the desired wavelength passes through the slit. The selected light passes into the sample compartment, through the sample and to the detector (photomultiplier). By measuring transmittance with (I) and without (Io) the sample present, absorbance can be determined. This is how a single beam spectrophotometer operates

2. **Double Beam Spectrophotometer**

This instrument has a chopper motor which alternately deflects the light beam through a reference cell and the sample cell. This is done many times per second, and the average ratio between the two readings gives the transmittance.

11: Principal of molecular hybridization

The technique of nucleic acid hybridization is established and developed on the basis of the denaturation and renaturation of nucleic acids. Hydrogen bonds in double- stranded nucleic acids can be disrupted by some physicochemical elements, and two strands of nucleic acids are separated into single strand. If different single-stranded DNA molecules, or DNA and RNA molecules, or RNA molecules are mixed together in a solution, and the renaturation is allowed to occur under proper conditions, single- stranded DNA or RNA will bind with each other to form a local or whole molecule of double-stranded structure as long as the single-stranded molecules are complementary, no matter what kind of sources they come from. Nucleic acid hybridization as a technique involves using a labeled nucleic acid probe, which is a known DNA or RNA fragment, to bind with the target nucleic acids, which is usually a poorly understood, heterogeneous population of nucleic acids. A probe labeled with detectable tracer is the prerequisite for determining a specific DNA sequence or gene in a sample or genomic DNA by nucleic acid hybridization. The target nucleic acids to be analyzed are usually denatured, and then mixed with the labeled probe in the hybridization system. The probe will bind to the segment of nucleic acid with complementary sequence under proper conditions. The hybridization can be identified by the detection of the tracer labeling the probe. Thus the existence or the expression of specific gene can be determined.

12: Note on contact freez dryer.

Contact freeze dryers use contact (conduction) of the food with the heating element to supply the sublimation energy. This type of freeze dryer is a basic model that is simple to set up for sample analysis. One of the major ways contact freeze dryers heat is with shelf-like platforms contacting the samples. The shelves play a major role as they behave like heat exchangers at different times of the freeze-drying process. They are connected to a silicone oil system that will remove heat energy during freezing and provide energy during drying times. • Additionally, the shelf-fluid system works to provide specific temperatures to the shelves during drying by pumping a fluid (usually silicone oil) at low pressure. The downside to this type of freeze dryer is that the heat is only transferred from the heating element to the side of the sample immediately touching the heater. This problem can be minimized by maximizing the surface area of the sample touching

the heating element by using a ribbed tray, slightly compressing the sample between two solid heated plates above and below, or compressing with a heated mesh from above and below.

13: How turtle preserved?

Preservative should be injected into the body cavity just in front of each of the four limbs, between the carapace and plastron. Use a long needle and continue injections until the head and · limbs are forced out of the shell. If a syringe is not available, make deep cuts into the body cavity just in front of each leg. Limbs, neck and tail should be injected or slit, as in large lizards.

Storage:

After the specimens have been injected or slit, tagged, and fixed, they should be put directly into preservative. If they are to be transferred later to plastic bags for temporary storage or to be shipped they should first be allowed to remain completely immersed in preservative for at least 48 hours if formalin is used, or a week if alcohol is used. The longer they are allowed to stay in preservative, the better. They should be loose and completely covered with plenty of liquid. Specimens which have been hardened in trays should also be allowed to soak in preservative for a day or two before being shipped or placed in plastic bags for storage.

If space is no problem, preserved specimens are best kept in glass containers. Bail-top jars with a glass top and rubber gasket are best. Fruit jars with a metal screw top lid may be used but should be carefully watched for rust and evaporation. Glass jars with polyethylene lids and liners are more commonly used in collections, since the lids form a tight seal and are easier to obtain than the traditional bail-top jars. Metal containers should be used only for temporary storage unless coated on the inside with paraffin, "Bakvar", or some other rustproof material.

14: Soxhelet extractor

A Soxhlet extractor is a piece of laboratory apparatus invented in 1879 by Franz von Soxhlet. • It was originally designed for the extraction of a lipid from a solid material. Typically, Soxhlet extraction is used when the desired compound has a limited solubility in a solvent, and the impurity is insoluble in that solvent. • It allows for unmonitored and unmanaged operation while efficiently recycling a small amount of solvent to dissolve a larger amount of material.

1. a percolator (boiler and reflux) which circulates the solvent,

2. a thimble (usually made of thick filter paper) which retains the solid to be extracted,

3. a siphon mechanism, which periodically empties the thimble.

4. The source material containing the compound to be extracted is placed inside the thimble.

5. The thimble is loaded into the main chamber of the Soxhlet extractor.

6. The extraction solvent to be used is placed in a distillation flask. The flask is placed on the heating element.

7. The Soxhlet extractor is placed atop the flask. 8. A reflux condenser is placed atop the extractor.

Working principle:

• The solvent vapors travels up a distillation arm and floods into the chamber housing the thimble of solid.

• The condenser ensures that any solvent vapors cool, and drips back down into the chamber housing the solid material. The chamber containing the solid material slowly fills with warm solvent.

• Some of the desired compound will then dissolve in the warm solvent.

• When the Soxhlet chamber is almost full, the chamber is automatically emptied by a siphon side arm, with the solvent running back down to the distillation flask.

• The thimble ensures that the rapid motion of the solvent does not transport any solid material to the still pot. This cycle may be allowed to repeat many times, over hours or days.

• During each cycle, a portion of the non-volatile compound dissolves in the solvent. After many cycles the desired compound is concentrated in the distillation flask.

• The advantage of this system is that instead of many portions of warm solvent being passed through the sample, just one batch of solvent is recycled.

• After extraction the solvent is removed, typically by means of a rotary evaporator, yielding the extracted compound.

• The non-soluble portion of the extracted solid remains in the thimble, and is usually discarded.

15: PCR, its types and uses:

Polymerase chain reaction (PCR) is a method widely used to rapidly make millions to billions of copies of a specific DNA sample, allowing scientists to take a very small sample of DNA and amplify it to a large enough amount to study in detail. PCR was invented in 1983 by the American biochemist Kary Mullis at Cetus Corporation. It is fundamental to much of genetic testing including analysis of ancient samples of DNA and identification of infectious agents. Using PCR, copies of very small amounts of DNA sequences are exponentially amplified in a series of cycles of temperature changes. PCR is now a common and often indispensable technique used in medical laboratory research for a broad variety of applications including biomedical research and criminal forensics. The majority of PCR methods rely on thermal cycling. Thermal cycling exposes reactants to repeated cycles of heating and cooling to permit different temperature-dependent reactions – specifically, DNA melting and enzyme-driven DNA replication. PCR employs two main reagents – primers (which are short single strand DNA fragments known as oligonucleotides that are a complementary sequence to the target DNA region) and a DNA polymerase. In the first step of PCR, the two strands of the DNA double helix are physically separated at a high temperature in a process called nucleic acid denaturation In the second step, the temperature is lowered and the primers bind to the complementary sequences of DNA. The two DNA strands then become templates for DNA polymerase to enzymatically assemble a new DNA strand from free nucleotides, the building blocks of DNA. MUHAMMAD IMRAN 97 As PCR progresses, the DNA generated is itself used as a template for replication, setting in motion a chain reaction in which the original DNA template is exponentially amplified.

Real-Time PCR (quantitative PCR or qPCR)

• Reverse-Transcriptase (RT-PCR)

• Multiplex PCR. • Nested PCR.

• High Fidelity PCR

. • Fast PCR. •

Hot Start PCR.

• GC-Rich PCR

16: Sanger sequencing

The DNA sequencing method developed by Fred Sanger from the basis of automated “cycle” sequencing reaction today. • In the 1980s two key developments allowed researchers to believe that sequencing the entire genome could be possible. • The first was a technique called polymerase chain reaction (PCR)that enabled many copies of DNA sequence to be quickly and accurately produced. • Sanger sequencing method of DNA sequencing was first commercialized by Applied Biosystems. • It was the most widely used sequencing method for approximately 40 years. • Regions of DNA up to about 900 base pairs in length are routinely sequenced using this method. • In the Human Genome Project, Sanger sequencing was used to determine the sequences of many relatively small fragments of human DNA. • The fragments were aligned based on overlapping portions to assemble the sequences of larger regions of DNA and, eventually, entire chromosomes.

Requirement:

• Requirements for Sanger sequencing Fred Sanger’s method of DNA sequencing was based on Arthur Kornberg’s earlier work on DNA Replication. A new DNA strand is synthesized using an existing strand as a template.Sanger sequencing involves making many copies of a target DNA region. They include: • A DNA polymerase enzyme • A primer, which is a short piece of single-stranded DNA that binds to the template DNA and acts as a "starter" for the polymerase • • The four DNA nucleotides (dATP, dTTP, dCTP, dGTP) MUHAMMAD IMRAN 10 5 • The template DNA to be sequenced However, a Sanger sequencing reaction also contains a unique ingredient: • Dideoxy, or chain-terminating, versions of all four nucleotides (ddATP, ddTTP, ddCTP, ddGTP), each labeled with a different color of dye.

METHOD

• To sequence DNA, four separate reactions are necessary – one to provide sequence information about each of the nucleotides. Each reaction contains: template DNA, a short primer (about 20 nucleotides), DNA polymerase, and the four dNTPs (one radioactively labeled). • One type of ddNTP – A, T, C, or G – is added to each. • Each reaction is then loaded into a separate lane of a polyacrylamide gel containing urea, which prevents the DNA strands from renaturing during electrophoesis. • Ionized phosphates give the DNA molecule a negative charge, so DNA migrate toward the positive pole of an electric field. The movement of DNA molecules through the polyacrylamide matrix is size dependent. • Over the course of electrophoresis, shorter DNA molecules will move further down the gel than larger ones. Millions of terminated molecules of the same size will migrate to the same place and “band” in the gel.

17: Four techniques to measure concentration...

Mass concentration

• The mass concentration pi is defined as the mass of a constituent of mi divided by the volume of the mixture V.

• pi= mi /V

• The SI unit is kg/m3 (equal to g/L)

Molar concentration

• The molar concentration cì is defined as the amount of a constituent ni (in moles) divided by the volume of the mixture V.

• ci= ni /V

• The SI unit is mol/m3 . However, more commonly the unit mol/L (= mol/dm3) is used

Number concentration

• The number concentration Ci is defined as the number of entities of a constituent Ni in a mixture divided by the volume of the mixture V.

• Ci= Ni / V The SI unit is 1/m3 .

Volume concentration :

The volume concentration(σi (not to be confused with volume fraction is defined as the volume of a constituent Vi divided by the volume of the mixture V.

• σi=Vi / V

• Being dimensionless, it is expressed as a number, e.g., 0.18 or 18%; its unit is 1.

• Volume Percent:

The volume percent is used to express the concentration of a solution when the volume of a solute and the volume of a solution is given

• Volume percent = [(volume of solute)/(volume of solution)] x 100%

18: :Explain analytical and preparative ultracentrifugation.

Analytical ultracentrifugation:-

The aim of Analytical ultracentrifugation is use to study molecular interactions between macromolecules or to analyse the properties of sedimenting particles such as their apparent molecular weight.

Preparative ultracentrifugation:-

The aim of Preparative ultracentrifugation to isolate and purify specific particles such as subcellular organelles.

19: Principal of southern blotting and how does it works?

Southern blot hybridization is an assay for sample DNA by DNA-DNA hybridization which detects target DNA fragments that have been size-fractionated by gel electrophoresis (Figure 4-1). In Southern blot hybridization, the target DNA is digested with restriction endonucleases, size-fractionated by agarose gel electrophoresis, denatured and transferred to a nitrocellulose or nylon membrane for hybridization. Southern blot hybridization technique is widely applied in researches since its invention. It could be applied for analysis of gene expression, screening of recombinant plasmids, analysis of gene mutation, and identification of the existence of a given DNA such as DNA from pathogenic microorganism. It could also be used to detect deletion of gene by restrictions mapping.

20: Multiplex PCR

Multiplex PCR is a variant of PCR method in which more than one target sequence is amplified using multiple sets of primers within a single PCR mixture. This enables amplification of several gene segments at the same time, instead of specific test runs for each. This technology was first used by Chamberlain et al. for the diagnosis of Duchenne muscular dystrophy (1988). Multiplex PCR is a space, time and cost-effective method for the genetic analyses that need to be repeated many times (e.g. sequencing). It requires a small amount of DNA (10–200 ng) as the starting template and can be performed on specimens with a suboptimal DNA quality. Though multiplex PCR has many benefits, optimization of it is equally challenging. While using multiple primer pairs, primers from one pair can interact with primers from another one. As each primer pair could have different requirements, there is not a single optimum melting temperature (Tm) and ΔG.

21: . Principle of weighting balance.

- The principle behind the operation of an analytical balance is known as “magnetic force restoration.” These balances utilize an electromagnet to measure the mass of an object indirectly. Instead of directly measuring the mass, they determine the force required to balance the sample and then use that force to calculate the mass.
- To achieve this, an analytical balance employs an electromagnet that generates a force to balance the mass of the sample placed on the measuring pan. By adjusting the strength of the electromagnet, the balance can accurately counteract the gravitational force exerted on the sample.
- The balance is equipped with a transparent enclosure or draft shield that surrounds the measuring pan. This enclosure has doors and serves the purpose of blocking external influences, such as dust and air currents, that could interfere with the weighing process. By maintaining a controlled environment within the enclosure, the accuracy and performance of the balance are preserved.
- When an object is placed on the measuring pan, the electromagnet generates a force to balance its mass. This force acts in the downward direction to counteract the gravitational force. The balance measures this force using an electromagnetic motor, which is sensitive to changes in the electrical current passing through it.
- The electrical current required to generate the necessary force is directly proportional to the weight of the object. By measuring this current, the balance can determine the mass of the object. The mass is then displayed on the screen of the balance, allowing the user to read the precise measurement.
- In some analytical balances, a “null detector” is incorporated. This detector uses a light source to indicate when the weight of the object is equal to the electromagnetic forces applied. When the weight is perfectly balanced, the null detector signals that the measurement is accurate.

22: Electron microscope:

Electron microscopy (EM) is a technique for obtaining high resolution images of biological and non-biological specimens. • It is used in biomedical research to investigate the detailed structure of tissues, cells, organelles and macromolecular complexes.• The high resolution of EM images results from the use of electrons (which have very short wavelengths) as the source of illuminating radiation. • Electron microscopy is used in conjunction with a variety of ancillary techniques (e.g. thin sectioning, immuno-labeling, negative staining) to answer specific questions. • EM images provide key information on the structural basis of cell function and of cell disease.

23: Limitations of Electron microscope:

electron microscopes have a practical resolution roughly 1,000 times better than the light microscope; with many electron microscopes, points closer than 0.5 nm can be distinguished, and the useful magnification is well over 100,000X.

24: phase contrast microscope:

Phase-contrast microscopy is particularly important in biology. • It reveals many cellular structures that are invisible with a Brightfield microscope • In Bright-field microscopy the Sample illumination is transmitted (i.e., illuminated from below and observed from above) white light and contrast in the sample is caused by absorbance of some of the transmitted light in dense areas of the sample • Bright-field microscopy typically has low contrast with most biological samples as few absorb light to a great extent. • Staining is often required to increase contrast, which prevents use on live cells in many situations. • Bright field illumination is useful for samples which have an intrinsic colour, for example chloroplasts in plant cells or that are stained. Annular filter in dark field microscope The phase contrast microscopy is a special adaptation of the light microscopy & helps to obtain a clear picture of living or unstained cells. •The adaptors convert minute difference in phase changes in transmitted light due to refractive indices of all cell organelles in to perceptible shades of grey •This allow organelles of the living cell to become visible with fair contrast in them.

25: Advantages of vibratome:

It cuts sections directly from fixed material, without embedding. 2. No PARAFFIN embedding required… No need to deparaffinise and rehydrate sections prior to immunostaining 3. Sometimes, samples need to preserve particular features that are lost during embedding in paraffin or epoxy resins. During embedding, samples undergo dehydration and heating, which may cause damage in those molecules or molecular regions that we are interested in. For example, to detect them with antibodies by immunohistochemistry or in situ hybridization. No need to dehydrate tissues prior to embedding, thus decreased loss of cell constituents 4. No high temperatures or harsh chemical treatments that may lead to antigen instability 5. Furthermore, some tissue or cellular features need to be studied in thick sections of about 100 µm or larger. For example, the morphology of neuron dendritic trees is difficult to image in **thin sections (4 to 10 µm), and we may also need to visualize dendrites after immune histochemistry.**

26: Numerical Aperture and its uses:

Numerical aperture is a number that expresses the ability of a lens to resolve fine details in an object being observed. • In MICROBIOLOGY, the numerical aperture of a microscope objective is a measure of its ability to gather light and resolve fine details of the specimen at a fixed object distance. light waves that are forming the Image pass through the specimen and enter the objective in an inverted cone as shown in fig 1 • Numerical aperture is concerned with the diameter of the objective lens in relation to its focal length. • Thus, it is related to the size of the lower aperture of the objective, through which light enters into it.

27: Common Fixative and its uses:

Commonly used fixatives are alcohol, formalin, glutaraldehyde, etc. • Formalin –routine • Glutaraldehyde – electron microscopy • Picric acid(Bouin's solution) – renal & testicular tissue • Alcohol (Carnoy's fixative) – cytologic smears, endometrial sampling • Osmium tetraoxide – CNS tissues & electron microscopy Method (Commonly used fixatives) change in pH, penetration of the fixative,}

Uses :Factor affecting fixation are temperature, preferred than the higher one}volume, time, etc. The lowest concentration of the fixative is 10% formalin or 2.5% glutaraldehyde is used.

28: Requirements of bright-field microscopy:

In Bright-field microscopy the Sample is illuminated from below and observed from above through the eye piece) • white light and contrast in the sample is caused by absorbance of some of the transmitted light in dense areas of the sample. • Staining is often required to increase contrast, which prevents use on live cells in many situations • Bright field illumination is useful for samples which have an intrinsic colour, for example chloroplasts in plant cells.

29: Applications/Uses of fluorescence microscopy:

• Fluorescence microscopy is a critical tool for academic and pharmaceutical research, pathology, and clinical medicine. • This method is used for demonstration of naturally occurring fluorescent material and other non- fluorescent substances or microorganisms after staining with some fluorescent dyes. e.g.; Mycobacterium tuberculosis, amyloid, lipids, elastic fibers etc. It helps to identify the specific molecules with the help of the fluorescence substances. • Tracing the location of a specific protein in the specimen. • Imaging structural components of small specimens, such as cells. • Conducting viability studies one cell Imaging the genetic material within a cell (DNA & RNA). • Viewing specific cells within a larger populations with techniques such as FISH. • View different proteins located in various portions of cells • The Fluorescence Microscopy allows the researchers to identify various different molecules in the targeted specimen or sample at the same time.

30: Principle of paper chromatography:

The principle involved can be partition chromatography or adsorption chromatography. Partition chromatography because the substances are partitioned or distributed between liquid phases. The two phases are water held in pores of the filter paper and the other phase is a mobile phase which passes through the paper. When the mobile phase moves, the separation of the mixture takes place. The compounds in the mixture separate themselves based on the differences in their affinity towards stationary and mobile phase solvents under the capillary action of pores in the paper. Adsorption chromatography between solid and liquid phases, wherein the solid surface of the paper is the stationary phase and the liquid phase is the mobile phase.

31: How Phase contrast microscope work:

• Regions of different composition likely to have different Refractive indices. Normally such differences cannot be detected by our eyes. However, PCM depicts them in terms of differences in intensity of brightness and darkness, which are visible to eye • In a Phase contrast microscope, one set of light rays comes directly from the light sources (surround waves – S waves). • The other set comes from light that is reflected or diffracted from a particular structure in the specimen (diffracted waves – d waves). • The images differences in refractive index of cellular structure. Light passes through thicker parts of cell is held up relative to the light that passes through thinner parts of cytoplasm • Light passing through one material & into another material of slightly different refractive index or thickness will undergo a change in phase. • This change is translated into variations in brightness of the structures. • Phase contrast is obtained with the help of the Phase ring by separating the central & direct ray from the diffracted ray. • The ring shaped illuminating light that passes the condenser annulus is focused on the specimen by the condenser • Some of the illuminating light is scattered by the specimen . • The remaining light is unaffected by the specimen and forms the background light. • Light that does not interact with the speciman is collected by the objective passes through the phase ring, and is regarded exactly ¼ wavelength. • The Phase shifted is not detectable by the eye so the resulting image on the image plane in the microscope appears as a normal bright background• If a microscope can delineate change of phase as a change in brightness or color, the eye or the camera, or photocell will be able to detect the microscopic areas causing the phase changes

32: Advantages of common fixation:

Fixation should be carried out as soon as possible • prevents autolysis and putrefaction. • Large pieces of tissues cannot be examined under the microscope for ongoing molecular or morphological alterations • Rapid and even penetration. • To preserve cells and tissues in a life like manner as possible. • Stabilize labile elements. • Must be rigid to allow sectioning. • Must allow staining. • Optical contrast must be induced for morphological examination. • Allow long storage of tissues.

33: Principle of compound microscope:

• Generally.. microscopes visualize an image by using a glass lens and magnification is determined by, the lens's ability to bend light and focus it on the specimen, which forms an image. • When a ray of light passes through one medium into another, the ray bends at the interface causing refraction. • The bending of light is determined by the refractive index, which is a measure of how great a substance slows the speed of light. The direction and magnitude of the bending of the light are determined by the refractive indexes of the two mediums that form the interface. • A medium with a lower refractive index such as glass to air, it normally speeds up the light penetration and making light bend away from the normal and when light is passed through a medium with a greater refractive index such as air to glass, it normally slows down and bends towards the normal, perpendicularly to the surface. • If an object is put between these two mediums i.e between water and air, in this case, a prism, the prism will bend the light at an angle. This is how the microscopic lenses work, they bend the light at an angle.

34: Separation Techniques:

• Separation techniques are those techniques that can be used to separate two different states of matter such as liquids and solids. • Separation processes or a separation method or simply a separation is methodology to attain any mass transfer phenomenon that convert a mixture of substances into two or more distinct product mixtures. • Separation is an important asset to purify component of interest from a mixtures.

35: Classification of filteration:

• Vacuum Filtration • Centrifugal filtration • Gravity filtration • Cold filtration • Hot filtration • Multilayer filtration

36: Applications of ion exchange chromatography:

• An important use of ion-exchange chromatography is in the routine analysis of amino acid mixtures. • The 20 principal amino acids from blood serum or from the hydrolysis of proteins are separated and used in clinical diagnosis. • Chelating resins are used to collect trace metals from seawater. • To analyze lunar rocks and rare trace elements on Earth.

37: What is confocal microscopy and applications of fluorescent microscope:

In this type of fluorescence microscope, high-resolution imaging of thick specimens (without physical sectioning) can be analyzed using fluorescent-labeled dye. 1) Intercalation of Et. br. causes a decrease in the negative charge of the ds DNA and an increase in both its stiffness and length. The rate of migration of the linear DNA-dye complex through gels is retarded by a factor of - 15%.

38: Parts of microscope

(i)Evepiece:

An eyepiece is that part of an optical system, which is directed to the viewer. It is a construction of at least one or more lenses. The function of the eyepiece in a microscope is to convert the realenlarged-intermediate-image from the objective into an enlargedvirtual-image. looks through to see the magnified specimen. The ocular lens has a magnification that can range from 5x to 30x, but 10x or 15x is the most common setting.

(ii) • three or four objective lenses, with magnification levels ranging Parts of microscope It contains the ocular lens, which the user

Objective lenses: It combines with the eyepiece lens to increase magnification levels. Microscopes generally feature 4x to 100x. Objective revolvers are used in microscopes with multiple objective lenses that have• different magnification factors. An objective (lens) is that part of an optical system, which is directed to the object. It´s• task is to collect the light rays that are reflected from the observed item. The objective generates a real-optical image.

39 : 1)Limitations….?

Very low contrast of most biological samples.• The practical limit to magnification with a light microscope is around 1300X. Although• higher magnifications are possible, it becomes increasingly difficult to maintain image Clearity as the magnification increases• • Low apparent OPTICAL RESOLUTION due to the material used. • Samples that are naturally colorless and transparent cannot be seen well, e.g. many types of mammalian cells. These samples often have to be stained before viewing. Samples that do have their own color can be seen without Preparation.

40: Homogeneous mixtures and heterogeneous?

mixtures. A homogeneous mixture is a mixture that is uniform throughout. A heterogeneous mixture is a mixture that is not uniform throughout, i.e., ingredients of the mixture are distributed unequally. Air is a homogeneous mixture of different gases, including oxygen, nitrogen, carbon dioxide, and water vapour.

41: Saw microtome and Laser microtome? Long

• It is especially used for hard materials such as teeth or bones. • The microtome of this type has a recessed rotating saw, which slices through the sample. • The minimal cut thickness is approximately 30 µm and can be made for comparatively large samples.

Laser microtome • It cuts tissue or other material with the help of photons instead of steel blades. • The method is contact-free and enables to cut tissue in its native state. • Special preparation techniques are not required. • Cutting by optical breakdown. • The thickness achievable is between 10 and 100 µm.

42: Multilayer filtration differ b/w hot and cold filtration? Long

This can refer to multiple layers of different material, including sand, gravel, or charcoal, where the different layers contain different particle sizes of that material. In this type of filtration, a mixture of liquid and insoluble solid particles is poured over the layers, and the solid particles are caught throughout, resulting in a filtered liquid.

Cold filtration: • Cold filtration makes use of very low temperatures, often by using an ice bath. Some substances, such as fatty acid particles, become suspended in the mixture as they cool down, which then allows us to filter them out more easily.

Hot filtration: • This is often used for crystalline compounds that contain impurities. The way this filtration is done is by melting down the crystalline compound, removing the impurities as the substance is still in liquid form, and finally recrystallizing the now pure substance. Often, it is recommended that the apparatus used in this filtration be heated up so that the filtered substance doesn't crystallize in the funnel and block the flow.

43: Advantages of SEM?

• The scanning electron microscope has many advantages over traditional microscopes. • The SEM has a large depth of field, which allows more of a specimen to be in focus at one time. • The SEM also has much higher resolution, so closely spaced specimens can be magnified at much higher levels. • Because the SEM uses electromagnets rather than lenses, the researcher has much more control in the degree of magnification. • Provides strikingly clear images, make the scanning electron microscope one of the most useful instruments in research today.

44: Cationic and anionic exchangers?

Cationic exchangers: • Cationic exchangers possess negatively charged group, and these will attract positively charged cations. These exchangers are also called "Acidic ion exchange" materials, because their negative charges result from the ionization of acidic group.

Anionic exchangers: Anionic exchangers have positively charged groups that will attract negatively charged anions. These are also called "Basic ion exchange" materials. Ion exchange chromatography is most often performed in the form of column chromatography. However, there are also thin layer chromatographic methods that work basically based on the principle of ion exchange.

45: Manual and automates Processing? Long

Manual Processing: Slow, most labor intensive method since transfer of specimens or changing reagents is done by hand. With an advance of automation this method is almost obsolete.

Automated Processing: Automated Processing Specimen-transfer or "dip and dunk" processors: instruments which transfer cassettes from station to station in a rotary or linear configuration. Fluid-transfer or "enclosed" instruments hold the specimens in a process chamber or retort and the reagents are pumped in and out during processing. Microwave assisted processing: might require manual transfer of specimen or reagents, it accelerates processing by heating reagents.

46: The rate of migration of DNA through agarose gels? Long

Factors affecting the rate of DNA migration Different factors determine the rate of migration of DNA through agarose gels.

Molecular Size of DNA: • Molecules of double stranded DNA migrate through gel matrices at rates that are inversely proportional to the log10 of the number of base pairs.

Conformation of the DNA: • Super helical circular (form I), • Nicked circular (form II), • Linear (form III) DNAs migrate through agarose gels at different rates • The relative mobilities of the three forms depend primarily • On the concentration and type of agarose • Strength of the applied current • The ionic strength of the buffer • Density of superhelical twists in the form I DNAEthidium Bromide: • Intercalation of Et. br. causes a decrease in the negative charge of the ds DNA and an increase in both its stiffness and length. • The rate of migration of the linear DNA-dye complex through gels is retarded by a factor of - 15%. **Voltage:** • At low voltages, the rate of migration of linear DNA fragments is proportional to the voltage applied. • However, as the strength of the electric field is raised, the mobility of high-molecularweight fragments increases differentially. • Thus, the effective range of separation in agarose gels decreases as the voltage is increased. • To obtain maximum resolution of DNA fragments >2 kb in size, agarose gels should be run at no more than 5-8 V/cm. 9)

47: Factors affecting resolution

d= λ/2 NA • The resolution of a microscope is a function of two factors as given below: 1. Numerical aperture (NA) of the optical components 2. Wavelength of light (λ)which is used to examine a specimen.

LIMIT OF RESOLUTION (LR): The minimum distance between any two closest points on the microscopic object, which can be resolved into two separate and distinct points on the enlarged image.

• d= λ/2 NA

Wavelength of light and RESOLUTION: • The smaller is the wavelength of light (λ), the greater is its ability to resolve the points on the object into distinctly visible finer details in the image

Thus, the smaller is the wavelength of light, the greater is its resolving power..

48: Bright field microscopy and light path? Long

Bright-field microscopy: • In Bright-field microscopy the Sample is illuminated from below and observed from above through the eye piece) • white light and contrast in the sample is caused by absorbance of some of the transmitted light in dense areas of the sample. • Staining is often required to increase contrast, which prevents use on live cells in many situations • Bright field illumination is useful for samples which have an intrinsic colour, for example chloroplasts in plant cells.

The light path consists of: 1. A trans illumination light source, commonly a halogen lamp in the microscope stand; 2. A condenser lens which focuses light from the light source onto the

sample; 3. Objective lens which collects light from the sample and magnifies the image. Ocularsand/or a camera to view the sample image.

49: TLC components? Long

• TLC system components consists of:

TLC plates, preferably ready made with a stationary phase: These are stable and chemically inert plates, where a thin layer of stationary phase is applied on its whole surface layer. The stationary phase on the plates is of uniform thickness and is in a fine particle size.

TLC chamber- This is used for the development of TLC plate. The chamber maintains a uniform environment inside for proper development of spots. It also prevents the evaporation of solvents, and keeps the process dust free.

Mobile phase- This comprises of a solvent or solvent mixture The mobile phase used should be particulate-free and of the highest purity for proper development of TLC spots. The solvents recommended are chemically inert with the sample, a stationary phase. A filter paper- This is moistened in the mobile phase, to be placed inside the chamber. This helps develop a uniform rise in a mobile phase over the length of the stationary phase.

50: Embedding?

Embedding is the process in which the tissues or the specimens are enclosed in a mass of the embedding medium using a mould. • Since the tissue blocks are very thin in thickness they need a supporting medium in which the tissue blocks are embedded. • This supporting medium is called embedding medium.

51: What is chromatography? Long

Chromatography is an important biophysical technique that enables the separation, identification, and purification of the components of a mixture for qualitative and quantitative analysis. A wide range of chromatographic procedures makes use of differences in size, binding affinities, charge, and other properties to separate materials. It is a powerful separation tool that is used in all branches of science and is often the only means of separating components from complex mixtures

52: Applications of Chromatography:

Pharmaceutical sector: • To identify and analyze samples for the presence of trace elements or chemicals. • Separation of compounds based on their molecular weight and element composition. • Detects the unknown compounds and purity of mixture. • In drug development.

Chemical industry: In testing water samples and also checks air quality.

Forensic Science: In forensic pathology and crime scene testing like analyzing blood and hair samples of crime place.

Molecular Biology Studies: Various hyphenated techniques in chromatography such as ECLC-MS are applied in the study of metabolomics and proteomics along with nucleic acid research. •HPLC is used in Protein Separation like Insulin Purification, Plasma Fractionation, and Enzyme Purification and also in various departments like Fuel Industry, biotechnology, and biochemical processes.

53: Condenser?

• Usually, the condenser is cooled to promote condensation of the vapor, which is collected. • The condenser ensures that any solvent vapors cool, and drips back down into the chamber

housing the solid material. The chamber containing the solid material slowly fills with warm solvent. • a condenser lens which focuses light from the light source onto the sample;

54: Sample Preparation for TEM?

• Since electrons are deflected by air molecules and are easily absorbed and scattered by solid matter, only extremely thin slices (20 to 100 nm) of a microbial specimen can be viewed in the average TEM. Such a thin slice cannot be cut unless the specimen has support of some kind; the necessary support is provided by plastic. • After fixation with chemicals like glutaraldehyde or osmium tetroxide to stabilize cell structure, the specimen is dehydrated with organic solvents (e.g., acetone or ethanol). Complete dehydration is essential because most plastics used for embedding are not water soluble. • Next the specimen is soaked in unpolymerized, liquid epoxy plastic until it is completely permeated, and then the plastic is hardened to form a solid block. • Thin sections are cut from this block with a glass or diamond knife using a special instrument called an ultramicrotome.

55: Random sampling

Random sampling is the purest form of probability sampling. Each member of the population has an equal and known chance of being selected. When there are very large populations, it is often ‘difficult’ to identify every member of the population, so the pool of available subjects becomes biased.

56: Collapse in lyophilization. Two points

• The point at which the product softens to the extent that it can no longer support its own structure. This can be a problem for many reasons: • • Loss of physical structure • Incomplete drying • Decreased solubility • Lots of ablation (splat)

57: Bacterial transformation in biology

- Isolation of CELL-FREE or NAKED DNA. The cells are broken and the DNA released. The cellfree DNA is subsequently isolated and collected.
- Mixing of Donor DNA with Recipient Competent Cells. The naked donor DNA is incubated with the competent recipient cells to which it binds.
- Uptake and Recombination of Donor DNA. Donor DNA binds to competent recipient cells, following which it enters the recipient celis. Portions of the donor DNA align, at random, with genes on the recipient DNA and segments of the two DNAs ore exchanged.

58: Components of rotory evaporator

• Rotary evaporator (or rotavap/rotovap) is a device used in chemical laboratories for the efficient and gentle removal of solvents from samples by evaporation. • When referenced in the chemistry research literature, description of the use of this technique and equipment may include the phrase "rotary evaporator", though use is often rather signaled by other language (e.g., "the sample was evaporated under reduced pressure"). • Rotary evaporators are also used in molecular cooking for the preparation of distillates and extracts. • A simple rotary evaporator system was invented by Lyman C. Craig. It was first commercialized by the Swiss company Büchi in 1957.In research the most common form is the 1L bench-top unit, whereas large scale (e.g., 20L-50L) versions are used in pilot plants in commercial chemical operations.

59: Uses or applications of western blotting

Western blot (Immunoblotting) : A technique for detecting specific proteins separated by electrophoresis by use of labeled antibodies

• The confirmatory HIV test employs a western blot to detect anti-HIV antibody in human serum sample. Proteins from known HIV-infected cells are separated and blotted on a membrane as

above. Then, the serum to be tested is applied in the primary antibody incubation step; free antibody is washed away, and a secondary anti-human antibody linked to an enzyme signal is added • The stained bands then indicate the proteins to which the patient's serum contains antibody. • Western blot can also be used as a confirmatory test for Hepatitis B infection.

60: Application of real time PCR?

Real-time PCR also called quantitative PCR (qPCR) is a variant of standard polymerase chain reaction in which amplification and simultaneous quantitation of a target DNA is done in the same PCR machine, using commercially available fluorescence-detecting thermocyclers. Fluorescent dyes specifically label DNA of interest and the amount of fluorescence generated is proportional to the quantity of DNA present. Depending on the available excitation source and detection filters, a variety of fluorescent dyes may be used in qPCR. Two most commonly use real-time PCR methods uses SYBR green (a dye that binds to double-stranded DNA but not to single-stranded DNA, and, when so bound, fluoresces) and TaqMan probes respectively.

61: Diagnostic application of PCR

Prospective parents can be tested for being genetic carriers, or their children might be tested for actually being affected by a disease. DNA samples for prenatal testing can be obtained by amniocentesis, chorionic villus sampling, or even by the analysis of rare fetal cells circulating in the mother's bloodstream. PCR analysis is also essential to preimplantation genetic diagnosis, where individual cells of a developing embryo are tested for mutations. • PCR can also be used as part of a sensitive test for tissue typing, vital to organ transplantation. As of 2008, there is even a proposal to replace the traditional antibody-based tests for blood type with PCR-based tests. • Many forms of cancer involve alterations to oncogenes. By using PCR-based tests to study these mutations, therapy regimens can sometimes be individually customized to a patient. PCR permits early diagnosis of malignant diseases such as leukemia and lymphomas, which is currently the highest-developed in cancer research and is already being used routinely. PCR assays can be performed directly on genomic DNA samples to detect translocation-specific malignant cells at a sensitivity that is at least 10,000 fold higher than that of other methods. PCR is very useful in the medical field since it allows for the isolation and amplification of tumor suppressors. Quantitative PCR for example, can be used to quantify and analyze single cells, as well as recognize DNA, mRNA and protein confirmations and combinations.

62: What is collapse and two reasons of it

• The point at which the product softens to the extent that it can no longer support its own structure. This can be a problem for many reasons: • • Loss of physical structure • Incomplete drying • Decreased solubility • Lots of ablation (splat)

63: What is Lyophilization

Lyophilization or freeze drying is a process in which water is removed from a product after it is frozen and placed under a vacuum, allowing the ice to change directly from solid to vapor without passing through a liquid phase. The process consists of three separate, unique, and interdependent processes; freezing, primary drying (sublimation), and secondary drying (desorption).

64: Enlist two most commonly used air-free techniques used in laboratory.

The two most common types of air-free technique involve the use of a glovebox and a Schlenk line.

Glovebox • The most straightforward type of air-free technique is the use of a glovebox. • A glove bag uses the same idea, but is usually a poorer substitute because it is more difficult to purge, and less well sealed. • Normal laboratory equipment can be set up and manipulated

through the use of the gloves. • By providing a sealed but recirculating atmosphere of the inert gas, few other precautions are necessary. • Inventive ways of accessing items beyond the reach of the gloves exist, such as the use of tongs and strings. • The main drawbacks to using a glovebox are the cost of the glovebox itself, and limited dexterity wearing the gloves. • Cross contamination of samples due to poor technique is also acute, especially where a glovebox is shared. • Because gloveboxes are expensive and have limited space, it is common for gloveboxes to be used to store, weigh, and transfer air-sensitive chemicals. • Reactions are thereafter carried out using Schlenk technique. The gloveboxes are thus only used for the most air-sensitive reactions.

Schlenk line • The other main types of techniques are associated with the use of a Schleck line. • The main methods here are: • counterflow additions, where air-stable reagents are added to the reaction vessel against a flow of inert gas. • the use of rubber septa with syringes to transfer liquids and solutions cannula transfer, where liquids or solutions of air-sensitive reagents are transferred • between different vessels stoppered with septa using cannulae. Liquid flow is achieved via vacuum or inert gas pressure. • Glassware are usually connected via tightly-fitting and greased ground glass joints. Round bends of glass tubing with ground glass joints may be used to adjust the orientation of various vessels. • Filtration under inert conditions poses a special challenge which is usually tackled with specialized glassware. • A Schlenk filter, consists of sintered glass funnel fitted with joints and stopcocks. • By fitting the pre-dried funnel and receiving flask to the reaction flask against a flow of nitrogen, • carefully inverting the set-up, and turning on the vacuum appropriately, the filtration may be accomplished with minimal exposure to air.